

Myles Barney

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RESEARCH INTERESTS

Machine Translation (MT) • Natural Language Processing (NLP) • Large Language Model Interpretability • Low-Resource Language Technologies • Multilingual NLP

EDUCATION

Brigham Young University

Ph.D. in Computer Science — Advisor: Dr. Steve Richardson

Started Fall 2025

Provo, UT

- 3-year department fellowship (\$119,000). Focus: Machine Translation, NLP, LLM Interpretability
- Planned coursework: Advanced Deep Learning, AI Model Interpretability, Reinforcement Learning, Evaluation Metrics for Language Models, Prosocial Language Models, Privacy

Brigham Young University

B.S. in Computer Science (Machine Learning emphasis), Minor in Mathematics

Apr 2025

Provo, UT

- GPA: 3.90/4.00 — Full-tuition Heritage Scholarship recipient
- Relevant coursework: Machine Learning, Deep Learning, Artificial Intelligence, Database Modeling, Linear Algebra, Probability & Statistics, Algorithms

PUBLICATIONS & MANUSCRIPTS

- The Mysterious Case of Neuron 1512: Injectable Realignment Architectures Reveal Internal Characteristics of Meta's Llama 2 Model. [Under Review]

RESEARCH EXPERIENCE

BYU MATRIX Lab — Research Assistant

Advisor: Dr. Steve Richardson — Machine Translation & Multilingual NLP

Dec 2023 – Present

Provo, UT

- Investigate LLM internal representations using injectable realignment architectures to probe neuron-level behavior in Meta's Llama 2, contributing to model interpretability research
- Develop language-family-informed training strategies for multilingual neural machine translation, improving translation quality for low-resource languages
- Design and run experiments using PyTorch and Hugging Face Transformers on multi-GPU clusters; perform data preprocessing, model fine-tuning, and ablation studies
- Author research manuscripts targeting top NLP venues (ACL, EMNLP, NAACL)

PROFESSIONAL EXPERIENCE

BYU Research Administration Office — Software Developer

Provo, UT

Dec 2023 – Feb 2024

- Built an NLP pipeline using Python, regex, and LLM-based classification to parse and categorize government grant application codes by acceptability
- Integrated the tool with email and a web server for automated processing, reducing manual review time for grant compliance staff

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, LaTeX

ML/NLP Frameworks: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, NLTK

Tools & Infrastructure: Git, Linux, Slurm (HPC), Weights & Biases, Docker, Jupyter, Pandas, NumPy

Natural Languages: English (native), Mandarin Chinese (professional working proficiency), Spanish (elementary)

SELECTED GRADUATE & ADVANCED COURSEWORK

Advanced Deep Learning (CS 674) • AI Model Interpretability (CS 601R) • Evaluation Metrics for Language Models (CS 501R) • Privacy in AI (CS 601R) • Database Modeling (CS 452) • Reinforcement Learning (CS 401R)

HONORS & ACTIVITIES

- BYU CS Department 3-Year PhD Fellowship (\$119,000), 2025–2028
- Heritage Scholarship (full tuition), Brigham Young University
- 8th place, 2023 Rocky Mountain ICPC Regional
- AP Scholar with Distinction
- Member, BYU Competitive Programming Association and Creators Club

VOLUNTEER EXPERIENCE

Chinese-Speaking Volunteer, Church of Jesus Christ of Latter-day Saints

Feb 2019 – Feb 2021

Taipei, Taiwan

- Developed professional Chinese proficiency through 2 years of full immersion in Taiwan
- Oversaw training logistics for 200+ volunteers; led weekly planning and goal-setting sessions